

## Team Application Exercises (tAPP-4)

**Instructions:** Work on problem 1 on your own to apply the concepts you learned in the pre-reading material to solve problems. Then discuss your code with your team members and present one solution as a team. Follow the same process for all the problems. Swap your team solution with another team for feedback.

### Task 1

Write a NumPy program to create an array of integers from 10 to 20. Write down its output when it is run.

### Task 2:

Write a program that creates a 3x3 array. The program should use `save_txt()` and `load_txt()` to save a given array to a text file and load it.

### Task3:

Perform the following operations: Addition, subtraction and matrix product:

| <u>array1</u>                               | <u>array2</u>                       |
|---------------------------------------------|-------------------------------------|
| [23, 45, 11]<br>[12, 23, 54]<br>[1, 23, 10] | [3, 5, 1]<br>[2, 3, 4]<br>[9, 1, 5] |

### Task4:

Perform the following operations: Addition, subtraction and matrix product:

| <u>array1</u>                       | <u>array2</u>                        |
|-------------------------------------|--------------------------------------|
| [3, 5, 1]<br>[2, 3, 5]<br>[1, 3, 1] | [8, 5, 1]<br>[2, 9, 4]<br>[2, 10, 5] |